

[P5] Supplement: finite residual bounds and equality exclusion

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Supplement to [P5], *Lower bounds for the Erdős–Lovász cover number* (`papers/paper_5_every_r.md`), 2 October 2026. Licence: CC BY 4.0.

This supplement keeps the finite arguments of earlier versions of [P5]. It is separate from the asymptotic theorem. The proposed $r \geq 16$ conclusion uses the finite classification in Lemma 13, which `code/verify_anchor_count.py` checks exhaustively.

Sections 2–4 and their lemma numbers retain the original numbering so that the finite proofs can be checked without changing their internal citations. The supplement uses the main manuscript’s residual setup and elementary bounds.

Setup

Use the residual system of Section 2 of the main manuscript: $q = 3t + u$, matched triple union W , unmatched set U , local graphs F_i and support S . Equations (1.1), (1.2), (2.1), (2.2) and Lemmas 1–2 refer to that manuscript. The finite lemmas numbered 3–13 below retain their original labels.

Peeling vertices of degree at least four selects k vertices and leaves a degree-three residual family J . Since $|E(H)| \geq q + 4k$ and $r \leq k + \tau(J)$, a bound $4\tau(J) \leq q + r + C_0$ gives $|E(H)| \geq 3r - C_0$.

2. Theorem 1: the sharp residual bound

Theorem 1 (proposed). *Every intersecting r -uniform family J with q hyperedges and maximum degree at most three satisfies $4\tau(J) \leq q + r + 3$.*

Corollary 1. $g(r) \geq 3r - 3$ for every $r \geq 1$.

Proof. Peeling and Theorem 1. $\square$

2.3 The odd equality case

By (1.2) and Lemma 2, if u is even then $4\tau \leq 4t + 2u = q + (t + u) \leq q + r + 2$. If u is odd, $4\tau \leq q + (t + u) + 2$. So Theorem 1 follows once *u odd implies $r \geq u + t - 1$* .

Suppose u is odd and, by Lemma 2, $r = u + t - 2$. Then $d = u - t - 3 \geq 0$, and (2.1) becomes

$$S \geq u(t + 2). \quad (2.3)$$

If $u \geq 11$, (2.3) and Lemma 1 give $t \geq u/2$. Then $d + 8 \leq u + 2$ and $12 \leq u + 2$, so every $S_i \leq u + 2$ and $u(t + 2) \leq t(u + 2)$, that is $u \leq t$, against $t \leq u - 3$.

For odd $u \leq 9$, the conditions $0 \leq t \leq u - 3$ and $u(t + 2) \leq t \max(12, u + 4)$ leave exactly

$$(u, t) \in \{(5, 2), (7, 3), (7, 4), (9, 5), (9, 6)\},$$

with $d = 0, 1, 0, 1, 0$ respectively.

Lemma 3 (small excess). *If every colour has $2e - s \leq 1$ and F has more than one colour, then $S_F \leq 12$, $S_F \neq 11$, and $S_F = 12$ only when F is a properly three-coloured K_4 . If also $u \leq 9$, the same holds when F has one colour.*

Proof. A connected component on $v \geq 2$ vertices has $2e - v \geq v - 2$, so each colour is a matching, or one path on three vertices plus a matching. A colour with a matching of size three leaves the other colours empty, so F has one colour and $S_F \leq u \leq 9$. A colour of support five is P_3 plus an edge, and the other colours are then confined to two edges, giving at most 9. Otherwise every support is at most four. Support four forces two disjoint edges, and two cross-intersecting two-edge matchings are two perfect matchings of one K_4 ; a third colour must be the third. $\square$

The cases $d = 1$. For $(9, 5)$, (2.3) asks $S \geq 63$ and Lemma 3 allows 60. For $(7, 3)$, (2.3) asks 35, so all three S_i are 12: three edge-disjoint copies of K_4 on seven vertices. Two such copies meet in one vertex and leave $K_{3,3}$ plus an isolated vertex, which holds no third K_4 .

The cases $d = 0$. Now every colour is a matching and $q = 2r + 1$, so (1.1) is an equality for every hyperedge: every vertex has degree three and any two hyperedges meet once. So every witness block is a triple and the F_i partition the edges of K_u . For $(5, 2)$, $qr = 55$ is not divisible by three. For $u \leq 9$ a local graph has at most six edges: one colour is a matching of at most four edges, and with several colours each is a matching of at most two, since a third disjoint edge would meet no other-colour edge. Six edges occur only as the three perfect matchings of a K_4 , and five only as K_4 minus an edge. For $(9, 6)$ the 36 edges of K_9 would split into six copies of K_4 , needing degree 8 divisible by three. For $(7, 4)$ the 21 edges of K_7 split into four parts of size at most six only as $6 + 6 + 6 + 3$, $6 + 6 + 5 + 4$ or $6 + 5 + 5 + 5$; two edge-disjoint K_4 leave $K_{3,3}$ plus a vertex, which excludes the first two. In the third, after one K_4 on four vertices, each copy of K_4 minus an edge contains a triangle and so takes one of the three edges among the other three vertices; it is then two of the four joined to two of the three, plus that edge. The three pairs among the three are distinct and meet, so edge-disjointness needs three disjoint pairs among four vertices, which is impossible.

This proves Theorem 1. $\square$

2.4 Sharpness, and degree four

For odd n take points $1, \dots, n$ and one vertex for each pair $([2])$. Then $q = n$, $r = n - 1$, a cover is an edge cover of K_n , and $\tau = (n + 1)/2$, so $4\tau = q + r + 3$.

Corollary 2. *If J has maximum degree at most four, then $4\tau(J) \leq q + r + 3$.*

Proof. Take a maximal matching of four-element blocks; their defining vertices cover $4a$ hyperedges and the rest has maximum degree three. $\tau \leq a + (q - 4a + r + 3)/4$. $\square$

Peeling at degree five then gives $|H| \geq (15r - q - 15)/4$ for the size q of the remainder, which improves the additive term in the corresponding step of [2] but not its coefficient.

3. The equality case

In Sections 3–4, suppose $|H| = 3r - 3$ and $\tau(H) = r$ with $r \geq 5$.

3.1 The parameters

If H had a vertex of degree at least five, peeling it first would give $|H| \geq |J| + 4(r - \tau(J)) + 1 \geq 3r - 2$ by Theorem 1. If every degree were at most three, $3r - 3 \leq 2r + 1$ would force $r \leq 4$. So the

maximum degree is four. (At $r = 3$ and $r = 4$ the bound $3r - 3$ is attained, and this is where the argument stops.)

Take a maximum matching of four-element blocks. Their $4a$ hyperedges form *groups* of four, and each group has its *group vertex*. Delete them and let J be the rest, of maximum degree three and $q = 3r - 3 - 4a$ hyperedges. The a group vertices and a cover of J cover H , so $\tau(J) \geq r - a$, while Theorem 1 gives $4\tau(J) \leq q + r + 3 = 4(r - a)$. So Theorem 1 is an equality for J , which by §2.3 forces u odd, $\tau(J) = r - a$, $r = u + t - 1$ and $q = 3t + u$. Consequently

$$u = 2a + 3, \quad t = r - 2a - 2, \quad d = u - t - 1 \geq 0, \quad r = 2u - 2 - d, \quad (3.1)$$

and (2.1) becomes $S \geq ut$. Write $H_0 = (u - 1)/2$.

The number $d = 2r - q + 1$ measures the slack in (1.1). When $d = 0$ every vertex of a hyperedge of J has degree three in J ; as d grows, vertices of degree two and one are allowed.

3.2 Local bounds

Lemma 4 (pointwise capacity). *For $x \in U$ let S_x be the number of hyperedges of W met by the witnesses at x . Then $S_x \geq t$. In one F_i , if x carries E edges in k colours, then $E - k \leq d$.*

Proof. The hyperedge x has exactly $u - 1$ witnesses, so exactly $r - (u - 1) = t$ other vertices. They meet at most $2t$ hyperedges of W , and all $3t$ must be met, so $3t - S_x \leq 2t$. The $u - 1$ witnesses at x reach S_x distinct hyperedges of W , so the number of witnesses “wasted” at x is $u - 1 - S_x \leq u - 1 - t = d$. Edges of one colour at x reach one hyperedge, so E edges in k colours waste $E - k$. $\square$

Lemma 5. *Every F_i with more than one colour has $S_i \leq \max(12, d + 8)$ and $S_i - e(F_i) \leq 6$. A one-coloured F_i has $S_i - e(F_i) \leq H_0$.*

Proof. The first case of Lemma 1 already gives $\max(12, d + 8)$. In the star case, F_i is a star at one vertex with E edges in $k \leq 3$ colours, and $S_i = E + k \leq d + 2k \leq d + 6$ by Lemma 4. For the second bound: with two disjoint edges of one colour on W , either every support lies in W and each colour has $s - e \leq 2$, or at most two other-colour edges exist, contributing at most two, while the first colour has $e \geq s - 2$. Otherwise every colour is a star or triangle with $s - e \leq 1$. A one-coloured graph has $s - e$ at most its number of components, at most H_0 . $\square$

3.3 Two augmentation bounds

Call a one-coloured F_i *large* if its matching number is at least four, and let m be their number.

Lemma 6. $2m \leq t$.

Proof. A large F_i uses one colour z_i ; its other two points of M_i are met by no witness. So every $x \in U$ must meet those $2m$ points with its t non-witness vertices. No such vertex meets two of them. Within one M_i , its triple and an edge of F_i avoiding x would replace M_i by two triples. Across M_i and M_j , an edge of F_i avoiding x and an edge of F_j avoiding x and those two endpoints, which matching number four provides, give three triples in place of two. Either way the matching was not maximum. $\square$

Lemma 7. $u \leq \max(12, d + 8)$.

Proof. Suppose not. By Lemma 5 every F_i with more than one colour has $S_i < u$, and a one-coloured F_i has $S_i \leq u$. Since $S \geq ut$, every F_i is one-coloured and spans U . Then $2e - u \leq d$ gives $e \leq (u + d)/2$, and its matching number is at least $u - e \geq (u - d)/2 > 4$. So $m = t$, and Lemma 6 forces $t = 0$, whereas $t = u - d - 1 \geq 8$. $\square$

3.4 The swap lemma

The groups come from a *maximum* matching of four-element blocks. That alone constrains every deleted hyperedge.

Lemma 8 (swap). *Let $A \neq A'$ be deleted hyperedges of the same group. If $w \in A$ and $w' \in A'$ are vertices of degree three in J , their sets of hyperedges of J meet.*

Proof. w lies in A and in three hyperedges of J , so it has degree four in H and its block is $\{A\} \cup T$ with T three hyperedges of J ; likewise $\{A'\} \cup T'$. Neither block meets another group. If $T \cap T' = \emptyset$, the two blocks are disjoint, and putting them in place of the group's own block gives $a + 1$ disjoint four-element blocks, against maximum cardinality. $\square$

Call such T a *triple* of A . Every vertex that a deleted hyperedge shares with a hyperedge of J is a vertex of J , so a deleted hyperedge meets J through triples and through vertices of degree two or one in J .

Proposition 1. *In the equality setting $d \neq 0$, and $d \neq 1$ when $r \geq 11$.*

Proof. Take a group; it has four hyperedges.

$d = 0$. By §2.3, $q = 2r + 1$ and every vertex of a hyperedge of J has degree three in J . So a deleted hyperedge A meets the $2r + 1$ hyperedges of J through at most $r - 1$ triples. Fix one, T . If the other at most $r - 2$ all met T , they would reach at most $3 + 2(r - 2) = 2r - 1$ hyperedges; so A has a triple T' disjoint from T . By Lemma 8 every triple of a group-mate A' meets both T and T' , so reaches at most one of the $2r - 5$ hyperedges outside them. With at most $r - 1$ triples, A' misses one, since $r - 1 < 2r - 5$ for $r \geq 5$.

$d = 1$. Now $q = 2r$, and (1.1) allows each hyperedge of J at most one vertex of degree two and none of degree one. So J has at most $q/2 = r$ vertices of degree two, each in at most two deleted hyperedges, since degrees in H are at most four: at most $2r$ incidences with deleted hyperedges. A deleted hyperedge whose triples pairwise meet reaches at most $1 + 2n_3$ hyperedges through n_3 triples and two through each other vertex, at most $1 + 2(r - 1) < 2r$ in all; so every deleted hyperedge has two disjoint triples. For a group-mate A' with n_2 vertices of degree two, the $2r - 6$ hyperedges outside those two triples are reached at most once per triple and twice per vertex of degree two, so $2r - 6 \leq (r - 1) + n_2$ and $n_2 \geq r - 5$. Every deleted hyperedge is such a group-mate, so the $4a \geq 4$ deleted hyperedges need at least $4(r - 5)$ incidences, more than $2r$ when $r \geq 11$. $\square$

4. Theorem 2: excluding equality from $r = 16$

Theorem 2 (proposed). $g(r) \geq 3r - 2$ for every $r \geq 16$.

4.1 The deleted hyperedges and their matching numbers

For each of the $4a$ deleted hyperedges A let L_A be the graph on U whose edge xy is present when $w_{xy} \in A$; write e_A , s_A , ν_A for its edges, support and matching number. Let $T = \sum_i e(F_i)$ and $D = S - T$.

A witness of degree two in J lies in at most two deleted hyperedges, one of degree three in at most one, since degrees in H are at most four; so $\sum_A e_A \leq u(u-1) - T$. Let $M = \sum_A (u - s_A)$. For $x \in U$ its t non-witness vertices have room for at most $3t$ other hyperedges of H ; they must meet the $3t - S_x$ hyperedges of W not met by witnesses, and every deleted hyperedge A whose graph misses x . So at most S_x deleted hyperedges miss x , and $M \leq S$. Using $\nu_A \geq s_A - e_A$ and $4a = 2u - 6$,

$$\sum_A \nu_A \geq u^2 - 5u - D. \quad (4.1)$$

Lemma 9. *Within one group there are no three disjoint edges from three different graphs L_A .*

Proof. Extend them to a matching of H_0 edges of the complete graph on U , leaving one point x . The witnesses of these edges cover U except x and meet three hyperedges of the group; one vertex of x that lies in the fourth hyperedge covers both. With the t defining vertices of W and the $a - 1$ group vertices of the other groups this is a cover of H of size $H_0 + 1 + t + a - 1 = r - 1$. $\square$

Lemma 10. *Three graphs each with a matching of size four and a fourth nonempty graph contain three disjoint edges from three different graphs.*

Lemma 11. *The same holds for four graphs with matching numbers at least 3, 3, 4, 4.*

Proof of both. Fix size-four matchings P, Q in two of the graphs and suppose no such triple exists. Call an edge e of another graph *blocked* if no two disjoint edges, one from P and one from Q , avoid it; every edge of the other graphs is blocked. The edges of P and of Q avoiding a blocked edge must cross-intersect. Each side keeps at least two, and one edge cannot meet three disjoint edges, so exactly two survive on each side and they form a four-cycle, which is a component of the union of P and Q , retaining multiplicity when an edge belongs to both matchings. Both endpoints of e lie in both supports: e must remove exactly two edges from each matching. Since the surviving four vertices already have their P and Q edges inside the cycle, no union edge leaves that component. The other graphs supply three disjoint blocked edges (a size-three matching for Lemma 11, a size-four matching for Lemma 10). Their six endpoints lie in both supports, and outside one such four-cycle there are at most four such vertices, so one of the other blocked edges touches it. Its surviving four-cycle cannot be the first cycle, because it avoids that blocked edge. Being a component of the same matching union, it is therefore a second disjoint four-cycle. So $P \cup Q$ is two disjoint four-cycles on the same eight vertices, and every blocked edge is a diagonal of one of them.

For Lemma 11 the two size-three matchings each consist of three of the four diagonals. The first has both diagonals of one cycle and the second at least one of them; take that one for the second, the other for the first, and an edge of P in the other cycle. For Lemma 10 the three size-four matchings are the three perfect matchings of each of the two copies of K_4 . Take an edge f of the fourth graph. If f removes two vertices of one copy, the edge on its other two vertices and an edge of a different matching in the other copy avoid f . If f removes at most one vertex of each copy, each copy keeps a triangle or all of K_4 , which contains all three matchings, so two edges of different matchings avoid f . In every case a rainbow triple exists. $\square$

Order the matching numbers of one group as $v_1 \leq v_2 \leq v_3 \leq v_4$. Greedy choice gives a rainbow triple when $v_2 \geq 1$, $v_3 \geq 3$ and $v_4 \geq 5$. With Lemmas 9–11, every group has $v_2 = 0$, or $v_3 \leq 2$, or all $v_i \leq 4$ with neither $v_1 \geq 1$ and $v_2 = v_3 = v_4 = 4$ nor $v_1, v_2 \geq 3$ and $v_3 = v_4 = 4$. So the sum is at most $\max(2H_0, H_0 + 6, 13)$, and with (4.1):

$$D \geq \frac{u^2 - 6u - 3}{2} \quad (u \geq 15), \quad D \geq 39 \quad (u = 13). \quad (4.2)$$

4.2 Large remainders: u is at most 11

$u \geq 15$. By Lemma 7, $d \geq u - 8$ and $t = u - d - 1 \leq 7$. By Lemma 5, $D \leq 7H_0 = 7(u - 1)/2$. With (4.2) this gives $u^2 - 13u + 4 \leq 0$, false for $u \geq 13$.

$u = 13$. Lemma 7 gives $d \geq 5$. If $d \geq 6$ then $t \leq 6$ and $D \leq 6t \leq 36$, against (4.2). If $d = 5$ then $t = 7$ and every $S_i \leq 13$, so $S \geq 91$ forces every $S_i = 13$. A one-coloured F_i then has $e \leq 9$ and matching number at least four, so it is large and $S_i - e \leq 6$. A multicoloured F_i cannot be the star case of Lemma 5, which gives at most $d + 6 = 11$, so $S_i = 13 = d + 8$ is the extremal case of Lemma 1: two other-colour edges pa , pb , and a first colour made of ab and six edges at p , so $S_i - e = 13 - 9 = 4$. Lemma 6 gives $m \leq 3$, so $D \leq 6 \cdot 3 + 4 \cdot 4 = 34$, against (4.2).

So $u \leq 11$.

4.3 u = 11: the anchor count

Call (x, c) *open* in a local graph F_j if no edge of colour c meets x and every edge of the other two colours does. Let A_j be the number of open pairs of F_j .

For a local graph F_i and $x \in U$, let P_x be the set of colours of F_i with no edge at x ; their points of M_i are not reached by the witnesses at x . Call a pair of colours in P_x *forbidden* if F_i has an edge of the third colour avoiding x , and call $c \in P_x$ *good* if, for every pair f of U that avoids x and is not an edge of F_i , some edge of F_i of a colour other than c avoids both x and f . For a set $R \subseteq P_x$ whose pairs are all forbidden, put g_x equal to the size of R , minus the number of colours of P_x outside R , minus the number of colours of R that are not good; take the best R , and let $G = \sum_x \max(g_x, 0)$, the *anchor value* of F_i .

Lemma 12 (demand). *If F_i has anchor value G , the other local graphs have at least $tu + G - S$ open pairs in all.*

Proof. Fix x and R . The t non-witness vertices of x must reach the $3t - S_x$ hyperedges of W that its witnesses miss, among them the points of R . No vertex reaches two points of R : its block and an edge of the third colour avoiding x would replace M_i by two triples. So $|R|$ distinct vertices reach the points of R , each reaching at most one further hyperedge, and the other $t - |R|$ reach at most two each. The $|R|$ vertices therefore reach at least $t + |R| - S_x$ further hyperedges missed by the witnesses, of which at most one per colour of P_x outside R lies in M_i .

Let $\{x, c, q'\}$ be such a vertex with $c \in R$ good and $q' \in M_j$, $j \neq i$. No edge of colour q' in F_j meets x , since the witnesses at x miss q' . If an edge f of F_j of another colour avoided x , goodness gives an edge e of F_i of a colour other than c avoiding x and f , and the blocks of $\{x, c, q'\}$, e and f are three disjoint triples in place of M_i and M_j . So (x, q') is open in F_j . Distinct q' give distinct pairs, and each colour of R that is not good spoils at most one. So at least $t + g_x - S_x$ open pairs come from x . Summing over the x with $g_x > 0$ and using $S_x \geq t$ for the rest gives $tu + G - S$. $\square$

Three anchors matter. A properly three-coloured K_4 on W has $G = 3(u - 4)$: for $x \notin W$ all three colours are missing, every pair is forbidden, and every colour is good, since the other two colours form a four-cycle on W and a pair f outside F_i meets W in at most one point. A spanning one-coloured large F_i has $G = 2u$: for every x its two unused colours are missing, forbidden together,

and good, since a matching of size four leaves an edge avoiding x and f . The graph Y_k , with edges pa and pb of two colours and a third colour made of ab and k edges at p , has $G = 2(u - 3)$.

Lemma 13 (finite classification; code/verify_anchor_count.py). *Let $u = 11$ and $1 \leq d \leq 4$. Every local graph with more than one colour has at most $12 - S_j$ open pairs, and every one with $S_j \geq 11$ has anchor value at least 16. A one-coloured graph of support s has $11 - s$ open pairs, plus two for each vertex common to all its edges.*

The one-coloured count is direct. The script checks every multicoloured local graph on eleven vertices that satisfies (2.2) and Lemma 4: those with $S_j \geq 11$ are the K_4 and Y_4 , and at $d = 4$ also Y_5 , with anchor values 21, 16 and 16.

Proposition 2. *In the equality setting, $u = 11$ with $1 \leq d \leq 4$ is impossible.*

Proof. Here $t = 10 - d$, every $S_j \leq 12$ by Lemma 5, and $S \geq 11t$. So $\sum_j (12 - S_j) \leq t \leq 9$ and some $S_j \geq 11$. That F_j is an anchor with $G \geq 16$: by Lemma 13 if it has several colours, and otherwise it spans U with at most $(11 + d)/2 \leq 7$ edges, hence matching number at least four and $G = 22$. By Lemma 12 the other local graphs carry at least $11t + 16 - S$ open pairs.

By Lemma 13 a local graph carries at most $12 - S_j$ open pairs, except a one-coloured one whose edges share a vertex. A star with k leaves has $12 - S_j = 11 - k$ and carries one more; a single edge or an empty graph would need $12 - S_j \geq 10 > t$. A star needs $11 - k \leq t$ and, by (2.2), $k \leq d + 1$, so $k = d + 1$ and it uses the whole budget; at most one occurs. So the local graphs carry at most $12t - S + 1$ open pairs, and $11t + 16 \leq 12t + 1$ would need $t \geq 15$. $\square$

4.4 Proof of Theorem 2

Suppose $r \geq 16$ and $|H| = 3r - 3$ with $\tau(H) = r$. By §4.2, $u \leq 11$, and $r = 2u - 2 - d$. If $u \leq 9$ then $r \leq 16 - d$, so $r = 16$, $u = 9$ and $d = 0$, excluded by Proposition 1. If $u = 11$ then $d = 20 - r \leq 4$: $d = 0$ is excluded by Proposition 1 and $1 \leq d \leq 4$ by Proposition 2. So $|H| \geq 3r - 2$ by Corollary 1. $\square$

4.5 The remaining finite case

At $r = 15$, the parameter choices are $(u, d, t) = (9, 1, 7)$ and $(11, 5, 5)$. Proposition 1 excludes the first. The second remains open: at $d = 5$, multicoloured stars can have support $u = 11$ with zero anchor value, and the supply estimate in Lemma 13 no longer applies. Thus the proposed equality-exclusion argument stops at $r \geq 16$ and does not settle $7 \leq r \leq 15$.

Additional finite consequence of stability

Corollary 3. *If the degree-four peeling of H leaves a remainder with $q \geq 2r - 3$, then $|H| \geq \lceil (28r - 67)/9 \rceil$.*

Proof. Here $d \leq 4$ and $L = 12$. For $u > 12$, (5.1) gives $\ell \geq (u - 21)/3$, and with $r = 2u - d - 2 + 3\ell$ this is $\ell \geq (r - 40)/9$. For $u \leq 12$, $r \leq 22 + 3\ell$ gives the same for $r \geq 13$, and for smaller r the bound is negative. So $B \geq (r - 40)/9$. For this peeling, set $E_0 = n - q - 4k \geq 0$ and $\xi = k + \tau(J) - r \geq 0$. The identity $n = 3r - 3 + E_0 + B + 4\xi$ gives the claim. $\square$

Theorem 3 and equation (5.1) in this additional consequence refer to Proposition 1 and equation (3.1) of the revised main manuscript.

Verification

Theorem 1 on every configuration with at most six hyperedges, and at low degree up to nine, and the sharpness family: `code/verify_residual_bound.py`. Lemmas 1, 3 and 5, the $d = 0$ structure of §2.3 and the $u = 13, d = 5$ step: `code/verify_local_graphs.py`. Lemmas 10 and 11 on 8, 9 and 10 vertices: `code/verify_rainbow_lemmas.py` and `code/verify_rainbow.c`. Lemma 13, the anchor values and the failure at $d = 5$ reported in §4.5: `code/verify_anchor_count.py`. The arithmetic of §2.3, §3.4, §4 and Corollary 3: `code/verify_p5_arithmetic.py`. The runs are recorded in `results/p5_checks.txt`. The non-finite steps have been read but not independently reviewed.

Acknowledgements and research status

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